

Graph Theory Presentation Guidelines

Discrete Mathematics - Spring 2026 Project

1. Title Slide

This slide should provide a clear and professional introduction to the project. Students must include the project title, course name, student names and IDs, instructor name, and submission date. The goal is to ensure immediate identification of the project context and team members.

2. Introduction to Graph Theory

In this section, students should introduce graph theory as a mathematical structure consisting of vertices (nodes) and edges (connections). The explanation should highlight its importance in modeling relationships in computer science, networks, and optimization problems.

3. Problem / Topic Definition

Students should define the selected problem or application that will be solved using graph theory. The implementation should clearly describe the real-world motivation and explain why graph-based modeling is necessary for solving the problem efficiently.

4. Representation of Graphs

This section should explain how graphs are represented in computing systems. Students must describe adjacency matrices, adjacency lists, and optionally edge lists. The implementation should show how graph data is stored and accessed in memory.

5. Method / Algorithm

Students should select a relevant graph algorithm (e.g., BFS, DFS, Dijkstra's Algorithm) and explain its working. The implementation should include step-by-step logic, describing how nodes are visited and how the algorithm solves the given problem.

6. Worked Example (Manual Solution)

In this section, students should manually solve a small graph example. The implementation should clearly show step-by-step calculations, demonstrating how the algorithm progresses from the starting node to the final result.

7. Implementation

Students must implement the chosen graph model or algorithm using any programming language (Python, C++, Java, etc.). The implementation should include well-structured code with comments explaining graph construction, processing logic, and algorithm execution.

8. Results

This section should present the output generated by the implemented program. Students should explain the final results clearly, such as shortest path, connectivity status, or traversal order, and include diagrams or screenshots if possible.

9. Discussion

Students should analyze the performance and limitations of the implemented approach. The discussion should include observations on efficiency, correctness, and possible improvements or alternative graph algorithms.

10. Real World Applications

This section should connect graph theory concepts to real-world systems. Students should explain at least 3–5 applications such as GPS navigation systems, social networks, communication routing, biological networks, and scheduling systems.

11. Conclusion

Students should summarize the entire project. The implementation should highlight the importance of graph theory in solving real-world problems and emphasize the effectiveness of the chosen algorithm or model.

12. References

Students should list all references used in IEEE or standard academic format. This may include textbooks, research papers, and credible online sources such as GeeksforGeeks or academic publications.